Effect of Mass Distribution on the Moment of Inertia of a Rotating Wheel

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Abstract

This experiment investigated how the radial distribution of mass affects the moment of inertia of a wheel in rotation. A known torque is applied to the wheel with the use of a falling mass, and the angular velocity was measured as a function of time using a light gate with an oscilloscope. Linear regression of the angular velocity data provided angular accelerations of $\alpha_{inner} = (42 \pm 1)$ rad s^{-2} and $\alpha_{outer} = (29 \pm 1)$ rad s^{-2}. These values were substituted into the rotational dynamics model to calculate the moments of inertia of $I_{inner} = (1.95 \pm 0.05) \times 10^{-4}$kg m^2 and $I_{outer} = (3.10 \pm 0.10) \times 10^{-4}$kg m^2. Repositioning the masses from inner to the outer radial positions increased the moment of inertia by approximately 56% (in a quadratic manner), in agreement with the relationship $I = \Sigma m_i r_i^2$, [1]. The results demonstrate that mass located further from the axis of rotation has a larger effect on the rotational inertia and validate the experimental method for measuring moments of inertia of rigid bodies.

1. Introduction

Rotational motion is crucial in many physical and engineering systems, including electric motors, turbines and flywheels [2,3,4]. A quantity known as the moment of inertia - which characterises a body's resistance to changes in angular velocity under applied torque - is fundamental for these systems. The moment of inertia depends on both the total mass and how mass is distributed relative to the axis of rotation. As a result, objects with similar mass and geometry can show different rotational behaviour if their mass distributions differ. Therefore, measuring and understanding this quantity is essential for predicting rotational acceleration for systems that require stable rotating machinery.

There are several experimental techniques for determining the moment of inertia. Kladivová and Mucha describe a method using a physical pendulum, in which the oscillation period of a rigid body is measured as a function of the pivot position and used to determine its rotational properties, including the moment of inertia [1]. This approach avoids the need to apply a known external torque using the falling mass and instead relies on gravitation restoring forces [1]. While effective these techniques require careful alignment and accurate timings [1].

The research question addressed in this investigation is: what is the effect of radial mass distribution on the moment of inertia of a rotating wheel? By the moment of inertia's definition, the larger the radial distance from the axis of rotation the greater the moment of inertia and therefore the lower the angular acceleration for a fixed applied torque. This investigation aims to test this relationship using a simple falling mass torque system and the use of a light gate to measure angular velocity as a function of time for two different configurations in mass distribution along the wheel.

In this paper, Section 2 identifies the theoretical framework describing rotational dynamics and the model used to calculate the moment of inertia from the angular acceleration. Section 3 describes the experimental apparatus, measurement procedure, and uncertainty estimation. Section 4 reports the angular velocity data, regression analysis, and the calculated moment of inertia for the two mass distributions. Finally, Section 5 compares the experimental results with theoretical expectations, outlines the limitations of the method, and summarises the main conclusions of the investigation.

2. Theory

Rotational motion of rigid bodies has laws that can be directly compared to those which govern linear motion. The rotational equivalent of Newton's second law relates the applied torque τ to the angular acceleration α using the moment of inertia

$$\tau = I\alpha \tag{1}$$

where the moment of inertia dictates a body's resistance to changes in angular velocity and depends on how the mass is distributed relative to the axis of rotation. Mass located further from the axis contributes more to the moment of inertia, so moving the mass outwards increases the rotational inertia even with the total mass being constant.

In this experiment, the torque applied to the wheel is generated using a small mass m_0 suspended from a thread wrapped around the wheel with radius R. Applying Newton's second law to the falling mass gives

$$m_0 g - T = m_0 a \tag{2}$$

Rearranging Equation (2) gives

$$T = m_0(g - a). \tag{3}$$

The tension produces a torque on the wheel

$$\tau = TR. \tag{4}$$

With the assumption that the thread doesn't slip and remains taut during the acceleration phase, the linear acceleration of the wheel has a relationship with the angular acceleration of the wheel by

$$a = R\alpha. \tag{5}$$

Substituting Equations (3) - (5), into equation (1) gives an equation for the moment of inertia with quantities that are measurable during the experiment

$$I = \frac{m_0(g - R\alpha)R}{\alpha}. \tag{6}$$

During the acceleration phase, frictional and air resistance are assumed negligible, and as a result the angular velocity is expected to linearly increase with time,

$$\omega(t) = \alpha t + \omega_0. \tag{7}$$

From the definition of moment of inertia, the configuration with the larger radial positions of masses results in a higher I. As the applied torque is fixed, Equation (1) predicts a smaller angular acceleration when the discrete masses are positioned further from the axis of rotation. This prediction is tested experimentally in this investigation.

3. Methods

The moment of inertia was determined by applying a known torque using a falling mass to a rotating wheel and measuring the angular velocity as a function of time during the acceleration phase of the rotation using a light gate connected to an oscilloscope. Linear regression of the angular velocity data yielded the angular acceleration, which was then substituted into Equation (6) to calculate the moment of inertia for different mass distributions.

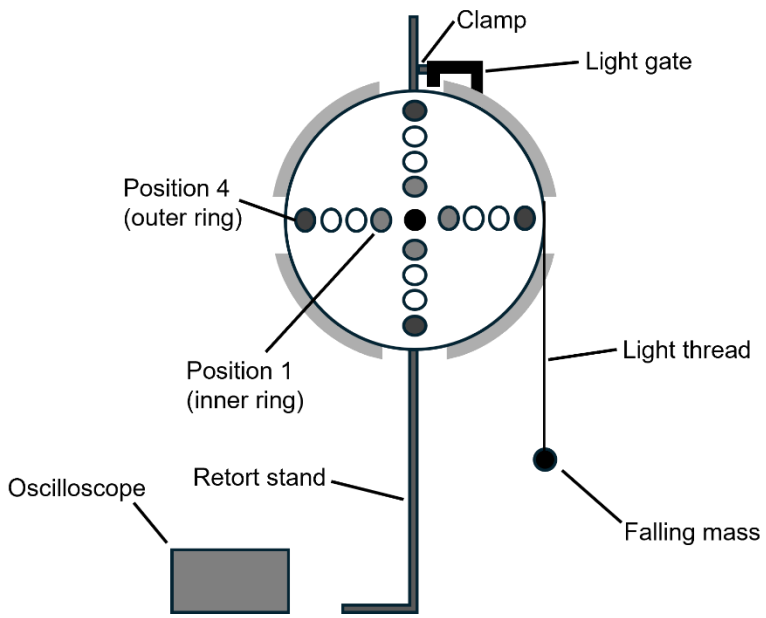

Figure 1: Schematic of the rotating wheel apparatus showing the radial slots used to position the metal dowels for the inner (position 1) and outer (position 4) mass configurations. A falling mass applies torque through the light thread, and a light gate is used to measure the rotational motion via the oscilloscope.

The experimental set up is shown in Figure 1. A rigid wheel was mounted on a low friction axle, which was held using a clamp and a retort stand. Light thread was wrapped around its circumference and attached to a small hanging mass. When released, the falling mass applied as tangential force on the wheel, generating a torque on the axis of rotation. The rotation was measured using a light gate positioned at the rim using the retort stand and a clamp, as shown in Figure 1. A notched paper disk was attached to the rear of the wheel, periodically interrupting the light beam, producing a square-wave signal, which was observed using a digital oscilloscope. The wheel radius was measured using a ruler and the falling mass using a digital balance. Metal dowels were inserted into the radial slots of the wheel, also shown in Figure 1, to vary the mass distribution whilst keeping the mass constant. Due to a temporary shortage in metal dowels, only four were used instead of the full set of eight, this isn't expected to influence the trends in the experiment.

Two configurations were investigated, with the masses placed on the inner radial position (position 1) and the outer radial position (position 4). For each mass configuration, the hanging mass was released from rest, the resulting square-wave signals being captured using the single-trigger mode of the oscilloscope. The width of each pulse at half maximum height was measured using the cursors on the oscilloscope, allowing the angular velocity to be calculated using $\omega = \frac{\Delta\theta}{\Delta t}$, where $\Delta\theta$ is the angular width of a notch and Δt is the measured pulse width. Two datasets of ten measurements each were recorded for every mass configuration.

Uncertainty in angular velocity was dominated by the uncertainty in the pulse-width measurement Δt made using the oscilloscope cursors. For the GW Instek GDS-1052-U, the manufacturer manual specifies a timebase accuracy of ±0.01% of the time [6]. Repeat measurements also showed differences in Δt, so the uncertainty in each ω data point was calculated from the standard error of the mean from repeat measurements. The uncertainty in angular acceleration was obtained from the linear regression of $\omega(t)$ and propagated to the calculated moment of inertia using partial differentiation with respect to α. The wheel's radius R was measured using a ruler, which had an uncertainty of ±0.05 mm, and the drop mass was assigned ±0.01 g based on the balance readability. The axle friction, air resistance, and slipping of the thread on the wheel posed as the systematic effects in the experiment, these were minimised through careful alignment of the axle and restricting analysis to the initial acceleration phase where frictional forces were small.

4. Results and Analysis

Angular velocity as a function of time was recorded during the acceleration of the rotating wheel. Two datasets were acquired, corresponding to the position of the masses inserted into the inner ring and outer ring positions on the wheel (position 1 and position 4, respectively). For each position, two sets of 10 measurements of angular velocity were taken and averaged, and the uncertainty was calculated using the standard error of the mean. The averaged data are presented in Figure 2, where linear relationships between angular velocity and time are observed for both mass distributions. The configuration with the inner rings (position 1) displays a steeper gradient, showing a greater angular acceleration in comparison to the outer ring configuration (position 4). No significant outliers were observed, although one data point was removed from the inner ring dataset because the wheel began to decelerate at the tenth pulse, which violated the constant acceleration assumption. The error bars show the uncertainty in angular velocity from repeat measurements, ranging from 0.3% to 31.7%.

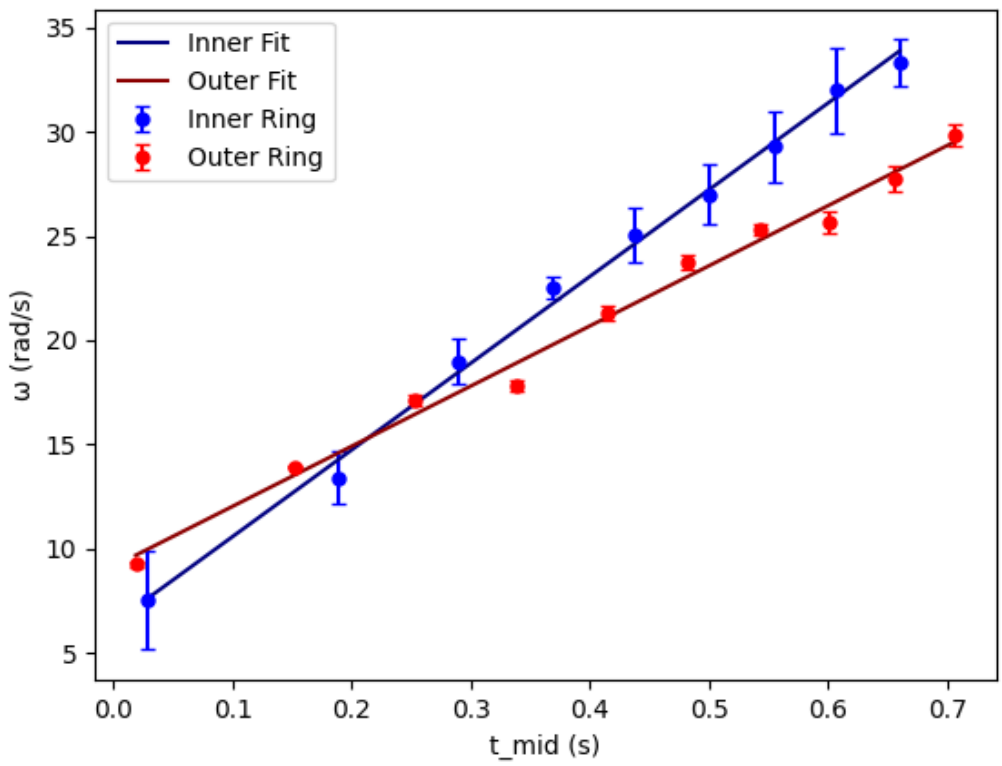

Figure 2: Angular velocity as a function of time for inner and outer mass distributions. Linear least squares fits are shown for both inner and outer ring data. Error bars were calculated using the standard error of the mean.

A linear least-squares regression was applied to each dataset, in accordance with the prediction of linear behaviour given by Equation (7) to determine the angular acceleration. The slope of each fit corresponds to the angular acceleration α. The fitted lines are included on Figure 2 for visual comparison. For the inner ring configuration, the regression exported $\alpha_{inner} = (42 \pm 1)$ rad s^{-2} with a coefficient of $R^2 = 0.998$. For the outer ring configuration, the results were $\alpha_{outer} = (29 \pm 1)$ rad s^{-2} with $R^2 = 0.995$.

To calculate the moment of inertia for each mass distribution, the angular accelerations were substituted into Equation (6). Using $m_0 = 0.02105$ kg for the falling mass, R = 0.050 m for the wheel radius, and g = 9.81 m s^{-2}, the values were $I_{inner} = (1.95 \pm 0.05) \times 10^{-4}$kg m^2 and $I_{outer} = (3.10 \pm 0.10) \times 10^{-4}$kg m^2. The difference between these values, $\Delta I = (1.10 \pm 0.10) \times 10^{-4}$kg m^2, corresponding to an increase of approximately 56% when the masses were relocated from the inner to the outer radial position. The uncertainty in I is dominated by the uncertainty in α obtained from regression, as discussed in Section 3. Uncertainty in I was calculated through propagating the regression uncertainty in α through Equation (6) using partial differentiation, with errors from m_0 and R being negligible in comparison.

5. Discussion and Conclusions

This investigation aimed to determine how redistributing mass within a rotating wheel affects its moment of inertia. A known torque was applied to the rotating wheel by a falling mass, the angular velocity was measured as a function of time using a light gate and oscilloscope, and linear regression was applied to obtain the angular acceleration for two mass configurations. These angular accelerations were then

substituted into the model developed in Section 2 to calculate the moments of inertia. Therefore, the objective of the experiment was to test whether relocating mass from inner to outer radial positions would lead to an increase in the moment of inertia, which was successfully achieved.

The angular accelerations for the two configurations, $\alpha_{inner} = (42 \pm 1)$ rad s^{-2} and $\alpha_{outer} = (29 \pm 1)$ rad s^{-2}, show a clear decrease in acceleration when the masses are positioned in the larger radii. Through Equation (6), this gave moments of inertia of $I_{inner} = (1.95 \pm 0.05) \times 10^{-4}$kg m^2 and $I_{outer} = (3.10 \pm 0.10) \times 10^{-4}$kg m^2, representing an increase of 56% for the outer configuration, as reported in section 4. This result is in direct agreement with the theoretical prediction from Equation (1) and Equation (7) that increasing the radial distance of mass with fixed applied torque increases I and therefore leads to a decrease in angular acceleration α. The high R^2 values confirm that the linear model of Equation (7) is suitable at describing the data during the acceleration phase.

The observed relationship between the moment of inertia and mass distribution is consistent with accepted results from mechanics. Equations such as $I = \Sigma m_i r_i^2$, show that relocating the mass to a larger radius increases I quadratically with distance from the rotational axis [1]. This is consistent in sign and order with the 56% increase in the moment of inertia.

Several limitations are present in this investigation which may have influenced the obtained values of I. Friction at the axle and air resistance was ignored in the theoretical model and these factors would reduce the angular acceleration, leading to a systematic overestimate of I. Manual measurement of pulse width using oscilloscope cursors introduces random uncertainty, and the relative uncertainty reached as high as 31.7% for some data points. Only four metal dowels were available rather than the full set of eight, reducing the size of the change in moment of inertia between configurations, although the relationship expressed is the same.

Future work could improve the precision of the method by automating the measurement of angular displacement. For example, Amrani describes a computerised rotational system where a data acquisition interface is used to record the rotational motion directly, removing the random uncertainty [7].

In conclusion, this experiment has demonstrated that relocating the mass further from the axis of rotation leads to a significant increase in the moment of inertia of a system in rotation. A change of $\Delta I = (1.10 \pm 0.10) \times 10^{-4}$kg m^2 was observed between the inner and outer configurations, corresponding to a 56% increase (see Section 4). This result is consistent with the predictions and experimental confirms that mass distribution strongly affects rotational inertia.

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